

Power System Analysis (NOC25-EE169)

Assignment-1 Solution

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 12

Total mark: 12 X 1 = 12

QUESTION 1:

Connected load = 200 MW

Maximum demand = 120 MW

MWhr generated in a year = 4.2×10^5

Average load = $\frac{4.2 \times 10^5}{8760} \text{ MW} = 47.945 \text{ MW}$

Load Factor = $\frac{\text{Average load}}{\text{Maximum demand}}$
= $\frac{47.945}{120} \approx 0.40$

Demand Factor = $\frac{\text{Maximum demand}}{\text{Connected load}}$
= $\frac{120}{200} = 0.60$

Correct Answer: a

QUESTION 2:

Plant rated capacity = 70 MW

Average load = $\frac{20 \times 6 + 30 \times 2 + 45 \times 4 + 60 \times 4 + 40 \times 4 + 50 \times 4}{24} = 40 \text{ MW}$.

Load Factor = $\frac{\text{Average load}}{\text{Maximum demand}}$

Maximum demand = 60 MW

Load factor = $\frac{40}{60} = 0.667$

Plant capacity factor = $\frac{\text{Average load}}{\text{Plant rated capacity}}$

Plant capacity factor = $\frac{40}{70} = 0.571$

Correct Answer: c

QUESTION 3:

Load on the power plant on a typical day:

Time	6AM – 8AM	8AM – 10AM	10AM – 2PM	2PM – 6PM	6PM – 6AM
Load (kW)	40	110	100	80	30

So, the peak load demand on the power plant = 110 kW

Sum of all individual max. load demands = (50+40+20+30) kW = 140 kW

Diversity factor = $\frac{\text{sum of all individual max. load demands}}{\text{Peak of all loads}}$

$$= \frac{140}{110} = 1.2727$$

$$\text{Coincident factor} = \frac{1}{\text{Diversity factor}} = \frac{1}{1.2727} = 0.7857$$

Correct Answer: c

QUESTION 4:

Energy supplied on each working day = [(80*4)+(40*6)] MWhr = 560 MWhr.

The power station operates for (365-50) = 315 days in a year.

So, energy supplied per year = (560*315) MWhr = 176400 MWhr.

$$\text{Annual load factor} = \frac{\text{MWhr supplied per annum}}{\text{Max. demand in MW} \times 8760} = \frac{176400}{80 \times 8760} = 0.2517$$

Correct Answer: b

QUESTION 5:

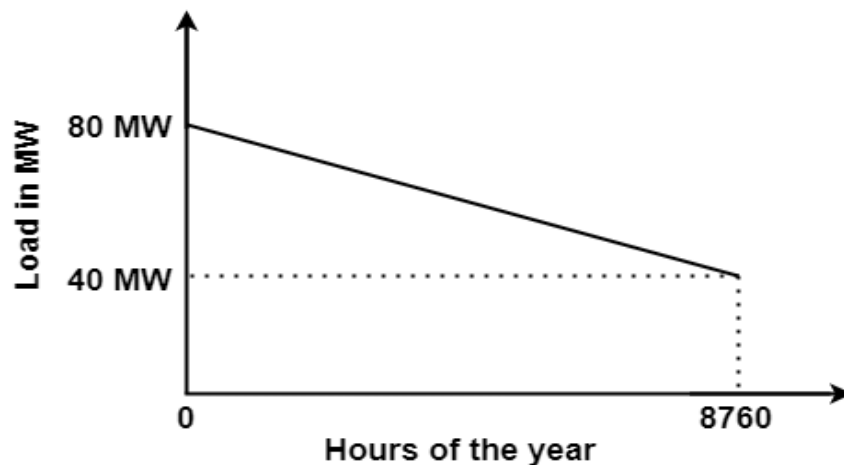


Fig.-Annual load duration curve

Installed plant capacity = (30*3+10*1) MW = 100 MW.

MWhr generated per annum = Area under annual load duration curve = $\frac{1}{2} \times (80 + 40) \times 8760 = 60 \times 8760$

So, average load demand = $\frac{\text{MWhr generated per annum}}{8760} = \frac{60 \times 8760}{8760} \text{ MW} = 60 \text{ MW}$

$$\text{Annual load factor} = \frac{\text{Average demand}}{\text{Peak demand}} = \frac{60}{80} = 0.75$$

$$\text{Plant capacity factor} = \frac{\text{Average demand}}{\text{Rated plant capacity}} = \frac{60}{100} = 0.60$$

Correct Answer: a

QUESTION 6:

Maximum demand = 90 MW

Load factor = 0.60

Plant capacity factor = 0.50

Plant use factor = 0.80

Load Factor = $\frac{\text{Average load}}{\text{Maximum demand}}$

Plant capacity factor = $\frac{\text{Average load}}{\text{Plant capacity}}$

$$\begin{aligned} \text{Utilization factor} &= \frac{\text{Maximum demand}}{\text{Plant capacity}} \\ &= \frac{\text{Plant capacity factor}}{\text{Load factor}} \\ &= \frac{0.50}{0.60} = 0.833 \end{aligned}$$

Correct Answer: d

QUESTION 7:

Maximum demand = 90 MW

Load factor = 0.60

Plant capacity factor = 0.50

Plant use factor = 0.80

Load factor = $\frac{\text{Average load}}{\text{Maximum demand}}$

So, Average load = Load factor X Maximum demand = (0.60 X 90) MW = 54 MW.

Rated capacity of the plant = $\frac{\text{Average load}}{\text{Plant capacity factor}} = \frac{54}{0.50}$ MW = 108 MW

Reserve capacity = Rated capacity – Maximum demand = (108-90) MW = 18 MW.

Correct Answer: c

QUESTION 8:

Maximum demand = 90 MW

Load factor = 0.60

Plant capacity factor = 0.50

Plant use factor = 0.80

Load factor = $\frac{\text{Average load}}{\text{Maximum demand}}$

So, Average load = Load factor X Maximum demand = (0.60 X 90) MW = 54 MW

Rated capacity of the plant = $\frac{\text{Average load}}{\text{Plant capacity factor}} = \frac{54}{0.5}$ MW = 108 MW

The maximum energy that could be produced daily if the plant operating schedule is fully loaded when in operation

$$= \frac{\text{Average load} \times 24 \text{ hr}}{\text{Plant use factor}}$$
$$= \frac{54 \times 24}{0.80} = 1620 \text{ MWhr}$$

So, number of hours of operation of the plant = $\frac{54 \times 24}{108 \times 0.80} = 15$

Therefore, the average number of hours during which the plant is not in service per day = 24-15 = 9.

Correct Answer: d

QUESTION 9:

The consequences of low power factor are increase in system kVA, increase in amount of copper in line, increase in power loss and poor voltage regulation.

Correct Answer: c

QUESTION 10:

Sum of maximum demands of different residential consumers = connected load $\times$ demand factor = (10 $\times$ 0.8) kW = 8 kW.

As the diversity factor among the residential consumers is 1.2, maximum residential load demand = $\frac{8.0}{1.2}$ kW = 6.667 kW.

Similarly, maximum commercial load demand = $\frac{12 \times 0.9}{1.1}$ kW = 9.818 kW

and maximum industrial load demand = $\frac{18 \times 1.0}{1.3}$ kW = 13.846 kW.

As the diversity factor among the different types of loads is 1.41, maximum demand on the feeder = $\frac{6.667+9.818+13.846}{1.41}$ kW = 21.511 kW.

Correct Answer: a

QUESTION 11:

Rate of annual load growth = g%

Years required (m) = 6

Initial load demand (P_o) = 80 MW

Final load demand (P_m) = 120 MW

Annual load after m years, $P_m = P_o \left(1 + \frac{g}{100}\right)^m$

or, $120 = 80 * \left(1 + \frac{g}{100}\right)^6$

or, $g = 6.99$

Correct Answer: b

QUESTION 12:

The relation between load factor (LF) and loss factor (LLF) is : $(LF)^2 < LLF < LF$

Correct Answer: b

*****END*****